FINASTRA

Lending

TRADE INNOVATION 2.11

SDK - Systems Integration Guide

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Version History

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Preface

The main purpose of this guide is to describe how Trade Innovation may interact with other systems:

- Behaving like a Trade Innovation zone
- Navigating into Trade Innovation from another application
- Navigating out of Trade Innovation to an external application

This is not an end-user task and so this document is targeted at technical consultants and bank staff. It assumes that you are familiar with the basic principles of trade finance and with Trade Innovation itself. Experience of programming is a requirement for understanding the more technical parts of the document.

The User Guide contains the following chapters:

Chapter	Description
Zone Proxy	Covers the expected behaviour of an application that will behave like a Trade Innovation zone.
URL Navigation into Trade Innovation	Covers the capabilities of navigation into a Trade Innovation zone.
URL Navigation from Trade Innovation	Covers the definition of external applications and navigation from Trade Innovation.

Screen Examples

This guide includes examples of screens. The screens shown in your system may not look the same as the examples in this guide. However the actual content and user operation are identical.

Document Conventions

Note: This style is used to indicate text that contains information of more than usual significance, such as warnings or hints in using the software.

Italicized font is used to distinguish terms, which have a specific meaning in the context of this document when they are introduced.

A `monospace` font is used to reference items such as file and folders names, artefacts, and code snippets.

Zone Proxy

A Zone Proxy is an application that will act like a Trade Innovation zone in regards to contributing data to the cross-zone dashboard, and providing navigation support to allow that data to be drilled into.

The situations this could be desirable are to:

- Access live transaction information from another system – a strategy for implementing Trade Innovation is to only add new transactions, allowing exiting transactions to be maintained and complete in the current systems – this facility would provide a combined view of outstanding transactions during that transition period
- Access historical data – another strategy is to take-on transactions in flight, but this would not consist of the full transaction history – so this can provide access to historical data
- Access to archived data – once data is archived from Trade Innovation there may be an occasional need to access that information.

The expectation is that a Zone Proxy is a light-weight web application that provides access to this legacy data. This document describes the two areas of behaviour that the Zone Proxy needs to support:

- Cross-zone Dashboard
- Cross-zone Navigation

Following is a technical description of the implementation of the cross-zone dashboard, and the requirements to support cross-zone navigation in order to facilitate the development of.

Cross-Zone Dashboard

The cross-zone dashboard aggregates information from multiple Trade Innovation zones. The central component to this is the cross-zone re-broadcaster which is part of the global application.

Dashboard Enquiry

The dashboard itself consists of three areas:

- Charts (for Supervisors)
- Work-In-Progress (WIP)
- Diary items

The dashboard is available in any zone, and will aggregate information across those zones the user is authorised to. The three areas of the dashboard screen correspond to three different sections of the message that drives the content gathering. Based on which refresh button is used on the screen indicates which parts of the request message are populated.

The dashboard also provides the ability to assign transactions to particular users/teams. The list of items currently shown in the WIP can all be re-allocated regardless of the zone each item originated from.

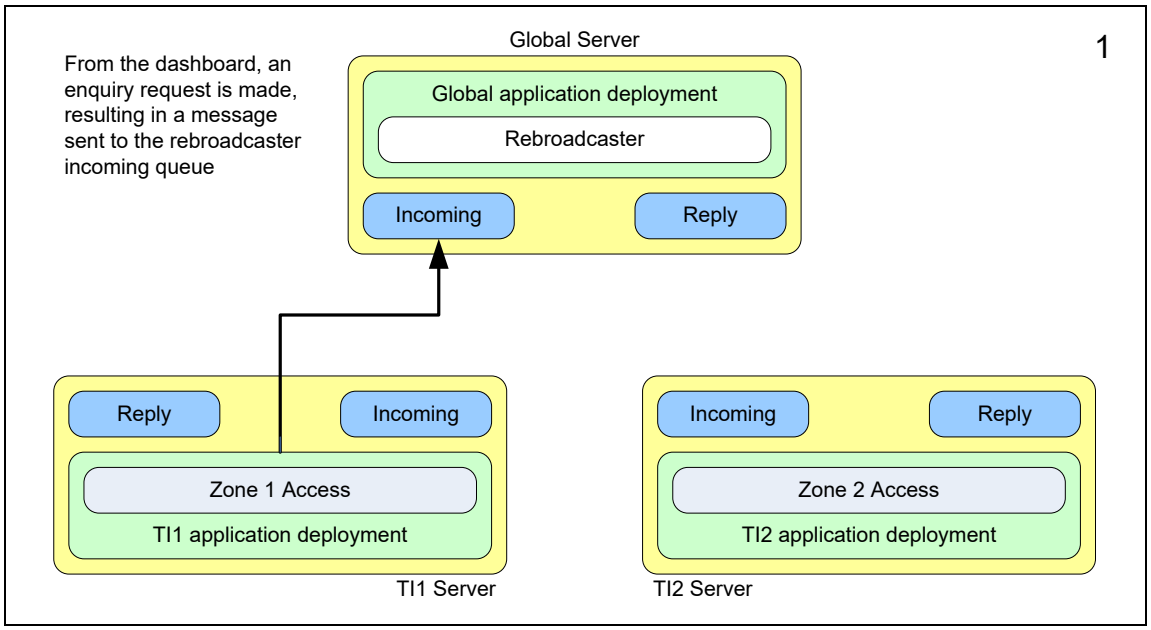
When selecting an item from the WIP, it is possible to view or continue the event, which means that if the item is in another zone the user is transferred to that zone with the event continuation being the first page shown.

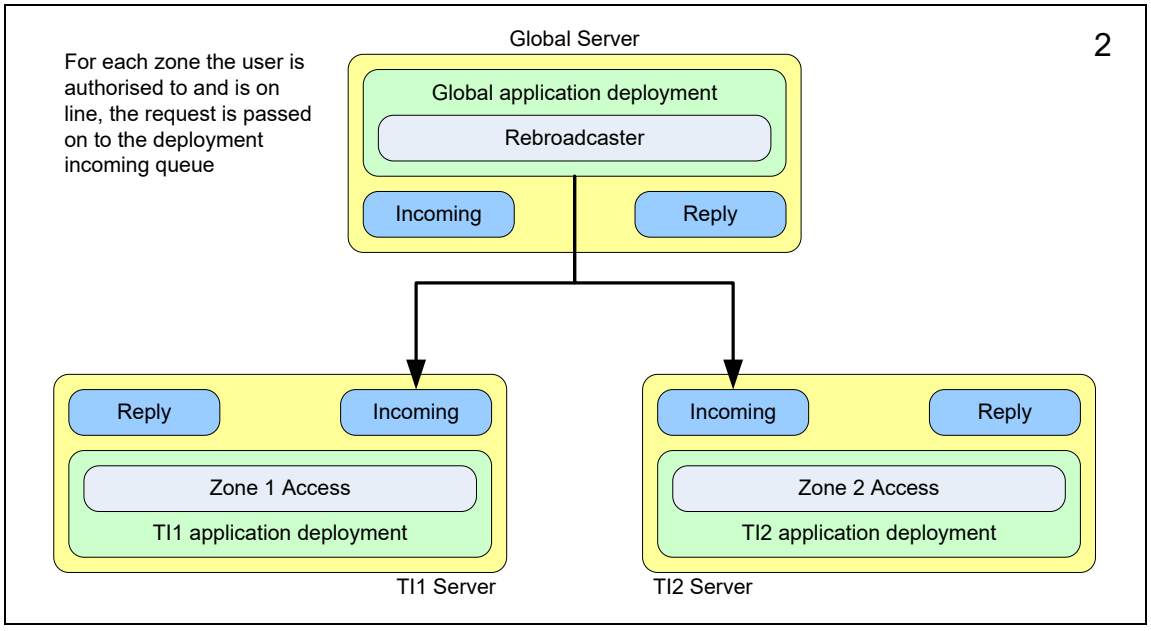
When selecting a diary item, it is possible to navigate to the master summary screen - again in another zone if appropriate.

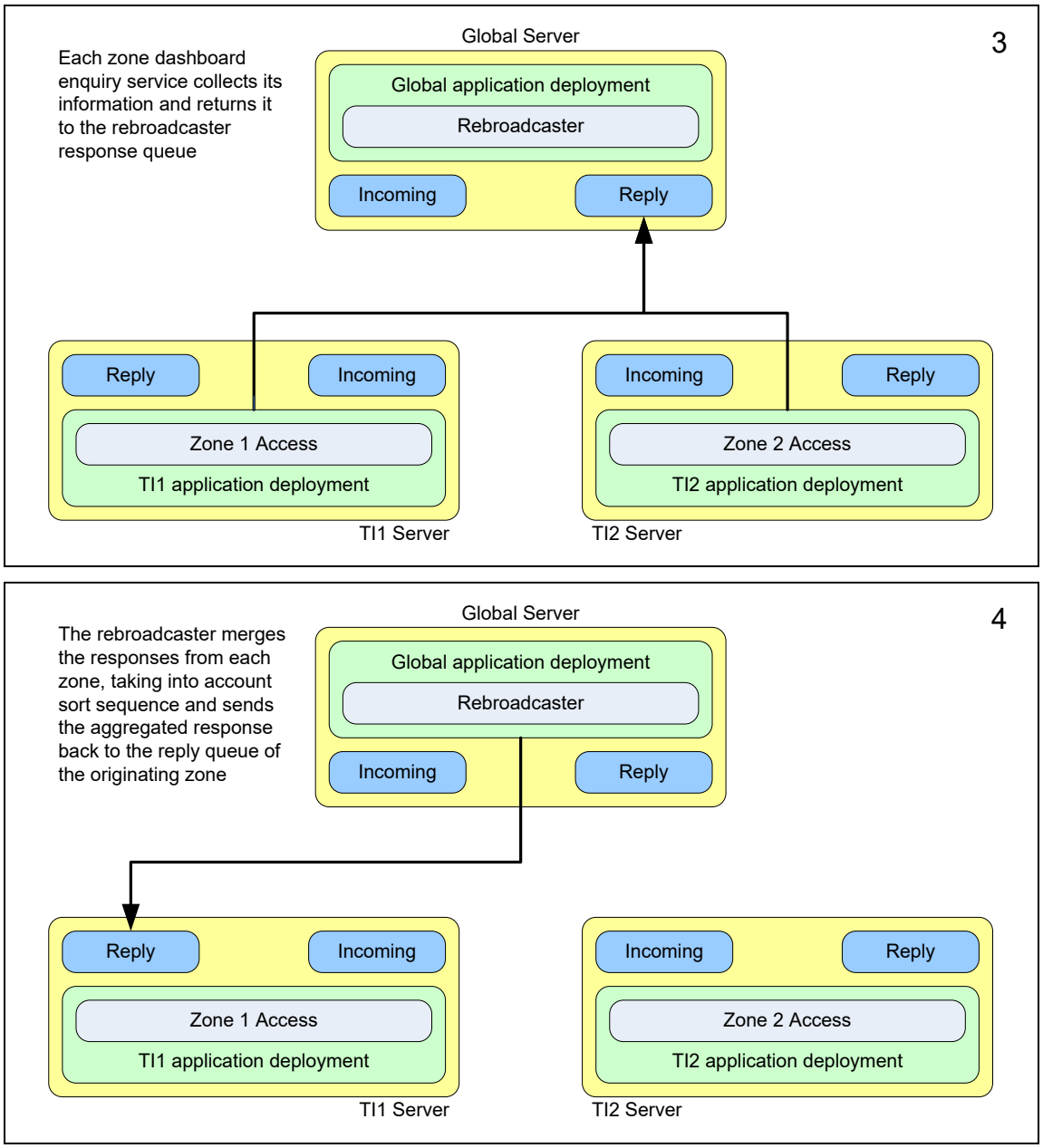
Finally it is also possible to transfer to the cross-zone dashboard in another zone as part of creating a new transaction.

Cross-Zone Rebroadcaster

The main component that backs this process is the rebroadcaster. The following sequence of diagrams illustrate the principles involved with the message handling used to gather the relevant data.







Two types of request are currently supported:

- relay request and aggregate response
- split request and aggregate response

The difference between them is the first will send the same request to all zones the user is authorised to that are online, whereas the second contains a set of zone-based requests that are split into zone-specific requests. This may be a subset of zones that the user is authorised to.

As responses are received, the data is aggregated using the original sort sequence requested. Each zone request is limited to return a maximum number of rows. This limit will also be applied to the aggregated response.

The use of this facility relies on a pseudo-synchronous JMS pattern. The JMS message that is sent to request the enquiry details will contain a correlation ID in the JMS header as well as a destination reply-to object. Once the messages have been sent to all of the relevant zones, the response queue is read repeatedly using the correlation id as a filter. The final response is merged as replies are received. Once all are received, the final response is returned to the original requester.

This process relies on a timeout on the response read – so that the user is not blocked forever if no response is returned. It also relies on the response messages having an expiry time associated with them, so if a timeout situation does occur, and then late-arriving responses expire on their own without a need for active cleanup from the application.

Message Processing

In order to provide content to the dashboard, you will need to configure an incoming JMS queue to your Zone Proxy that could receive the following messages:

- TI.ZoneDashboardRequest
- TI.ZoneReallocateTransactionsRequest

For the first message, the request will contain the content of the enquiry filters required to subset the information for return. A sort sequence and a maximum number of responses are also provided. The response is a `TI.ZoneDashboardResponse` message that contains the data corresponding to the filters passed.

Note: Note that the content of the enquiry controls the SLA behaviour - not the dashboard, so the calculation of the return data must take this into account.

For the second message, the request will contain a list of transactions to reallocate to a particular team or user. If this functionality is not to be supported, then default responses may be returned. The response is a `TI.ZoneReallocateTransactionsResponse` message, which contains the same details as requested, except updated to indicate that the reallocation has occurred (this data is used to update the dashboard enquiry rather than the content being re-requested).

The event/master level navigation relies on passing the key value as a single integer. This may need to be generated based on the data available both for the enquiry, and for any subsequent navigation.

The response messages need to be sent back to the rebroadcaster incoming queue, so that also needs to be visible to your application - though the original JMS request message will have the JMSReplyTo header field set, so no JNDI lookups need to be made. When creating the response, make sure that the JMSCorrelationId header field from the request is populated in your response.

Finally, the infrastructure does not wait for replies indefinitely, and will provide information to the user where a zone's response has timed out. In order to prevent those responses that are finally produced not to hang around on the reply queues, the connection factory, or the JMS message sender must specify the time-to-live parameter. With this value set to a time period (the same value as the timeout period would be appropriate); the responses that were no read will be automatically dropped from the reply queues after that period of time.

Cross-Zone Navigation

Cross-zone navigation covers the following cases:

- Continuing an event
- Viewing master summary
- Dashboard to Dashboard in order to create a master
- Direct navigation from the global application

If the first option is taken, the event screen is shown, and once the user exits from that, the master summary is shown. If the user then exits from the master summary, then the dashboard in that zone is shown.

In the first three cases, the filters defined for the dashboard from which the navigation originated are also passed so that when the dashboard in the target zone is shown, the same filters are applied, and so the same transactions are shown.

The redirection transfers through the global application, as that contains the URL for the target zone. The redirection simply strips the parameters from the incoming URL, and attaches them to the target zone URL.

The fields passed on the URL are:

Field	Description
pane	Entry point: • `TIPlusWP` - application list • TIPlusEventRedirectWP - continue event • `TIPlusMasterRedirectWP` - view master summary • TIPlusProcessRedirect - main dashboard • TIPlusCleanPaymentRedirectWP - clean payments dashboard • TIPlusFactoringRedirectWP - factoring dashboard
key	Raw internal id of master or event.
zoneTransferFilter_xxxxx	Filter value for the field xxxxx.
sourceZone	Zone from which the navigation originated.
zone	Zone id of target.

The Zone Proxy application should provide the means to at least view the equivalent of a master or event if the pane parameter passed is `TIPlusEventRedirectWP` or `TIPlusMasterRedirect`.

If the pane parameters are any of the three dashboard values, and you are not to intending to provide that level of functionality, then simply provide a means to navigate back to the originating zone.

To provide a means of returning to the originating zone, construct a URL starting with the URL of the global application, and add the parameters `pane=TIPlusProcessRedirect`, `zone=<sourceZone>` and provide the `zoneTransferFilter` values originally received.

If the Zone Proxy is navigated to directly from the global application, then the URL used will just be the URL configured for the Zone Proxy. In this case, provide a means to return to the global application.

The pane value of `TIPlusWP` is only achieved by externally accessing a Trade Innovation zone, and so should be treated in a similar way to the user navigating to the global application. In these cases, the URL may contain a `source` parameter that provides the URL to return to, in which case, use that over the global application URL to provide a means to navigate back.

Configuring a Zone Proxy

A Zone Proxy may be configured through the global application.

Define a new deployment from the Deployment Administration menu option. Provide the URL to the application that represents the Zone Proxy, and select the checkbox indicating this is an external deployment. You can then define a zone that uses that deployment from the Zone Administration menu option.

Users can then be authorised to the Zone Proxy in the same way as they would be a standard Trade Innovation zone.

When a user logs on to the global application, if they are authorised to a Zone Proxy, then it will appear in their list of zones with a status of N/A.

Security

The security associated with the Zone Proxy should be compatible with the mechanism configured for the Trade Innovation application, to prevent an unnecessary number of authentication steps that a user may be presented with.

URL Navigation into Trade Innovation

URL navigation into a Trade Innovation zone is available to:

- View or continue an event
- View the master summary page for a particular master

The behaviour of this navigation is a transfer into the zone, rather than for an external application to open a new window using this navigation, as complications can arise if multiple windows are opened, or if the same window is used and a new URL is used. The user's session in the Trade Innovation window(s) would not be in a compatible state.

URL Construction

The construction of the URL to use starts with the URL of the global application. To this you can add the following parameters:

Parameter	Description
`zone`	The identifier of the zone to navigate to.
`masterRef`	The master reference of the transaction.
`eventRef`	The event reference of the transaction – only required for event viewing or continuation.
`stepId`	The identifier of the step the event is currently at (only needed if parallel steps are in progress – otherwise master summary will be shown).
`sessionType`	Set to view or continue.
`pane`	Set to `ExternalSystemAccessWP`.
`source`	URL to use to return to the originating application.

Based on the parameters passed, the page opened will either be the master summary page or the event page. If there is a problem with the data, an error will be shown with a button to enable the return to the external application.

The master summary page will have a menu option of **Return** rather than **Close**, which will return to the external application using the source URL.

Note: Note that the source URL is not actually rendered on the web page behind the menu option, it is stored in the user's session and will be used to navigate to on demand.

When Trade Innovation returns to the external application, the user's session is closed.

Security Considerations

Both applications should be controlled by a common authentication facility. While Trade Innovation will prompt the user to log in if they are not already authenticated, a more seamless process is desirable.

URL Navigation from Trade Innovation

If a user spends most of their time in Trade Innovation, but occasionally need to access an alternate application to perform self-contained tasks, then it is possible to add entries to their application list representing those applications.

For this feature to work, the external application should have a similar feature of navigating back to where the request came from.

External Application Definition

External applications can be defined from the Trade Innovation zone Security application. When defining an external application, you provide an identifier, description and URL. The URL can contain static parameters.

When defined, a capability is generated. This can then be assigned to the appropriate users.

When you delete an application, the capability is removed as well as the user capability maps.

Selecting an External Application

If the user is authorised to an external application, then its description will appear in the application list when they access the zone it is defined in.

When the user selects the application and presses **Start**, the URL defined for the application is navigated to with the following parameters added:

Parameter	Description
zone	The identifier of the zone where the navigation originated.

As part of this navigation, the user's session is closed.

For the external application to return to the Trade Innovation zone, the URL required is the URL of the global application with a parameter of `zone` set to the respective zone id.

Security Considerations

Both applications should be controlled by a common authentication facility. While Trade Innovation will prompt the user to log in on return if they are not already authenticated, a more seamless process is desirable, especially if the external application will also prompt the user to authenticate.



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